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"The study of the military, the study of war, the study of warfare," the monograph | How to develop the concept of warfare

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How the operational concept was actually developed

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Introduction

The operational concept is an important theoretical grip for forward-looking design of future warfare and leading military construction and preparation, which originates from war preparation and is tested by war practice. In essence, the concept of operations development is based on the evolution of warfare, frontier technology development, combat opponents and changes in the counter environment, for strategic, operational, tactical different levels of operational problems, the design of methods and strategies and capabilities to support the optimization of the program, following the basic paradigm of "operational problems + methods and strategies + capabilities to support".

"Three layers and three segments" operational concept development architecture

Operational concept development is a systematic project. According to the level of the operational problem, the operational concept can be divided into three levels: top-level concept, operational concept, and support concept, and each concept design goes through three stages of operational problem analysis, method and strategy design, and capability support demonstration, i.e., the "three levels and three stages" operational concept development architecture.

From a hierarchical point of view, the top-level concept mainly plans and designs the warfare concept and overall warfare method for maintaining national strategic security and expanding interests; the operational concept mainly outlines and designs the operational concept and campaign warfare method for specific operational directions and fields; and the supporting concept mainly refines and designs the engagement scenarios and tactical procedures for conducting combat operations.

From the stage, in the operational problem identification phase, the main threat challenges are examined and the list of operational problems is sorted out. The focus is on standardizing the description of operational space-time, operational adversaries, operational threats, environmental posture, mission tasks, and other content to form the background elements of the operational concept. In the methodological strategy design phase, focus on the list of operational problems and design the preferred solution to the problem. The focus is on the operational concept, the standardized description of the connotation of the key meaning, the mechanism of victory, combat scenarios, operational methods of warfare, interaction, support conditions, etc., to form the operational concept of the elements of use. In the demonstration phase of capability support, for the preferred option, mapping analysis of the ability to support the needs of combat operations. The focus is to standardize the description of construction needs, capability indicators, ways to achieve, etc., to form the support elements of the operational concept.

Adhere to the problem-oriented, to identify the largest combat pain points

The concept of warfare is the starting point for the development of the concept of warfare. The core work of operational problem analysis is to adhere to the targeted problem-oriented, accurate research and judgment of the objective existence of the main threat, systematic analysis of the core tasks undertaken by the main combatant, thorough sorting of the key issues that need to be addressed, the direction of the development of the operational concept off.

Threat Research. Combining the military strategic guidance, combat force deployment and frontier technology development of both sides of the confrontation, we study and analyze the threats and challenges faced by different levels of military subjects in carrying out their missions according to the conceptual scenarios of strategic games and military confrontation. For the

strategic level, based on the opponent's national security strategy, national defense policy, military strategy, etc., the strategic threats to the expansion of the national security and interests of the own side in the long term are analyzed. For the battle level, based on the strategic game and military confrontation trends, combined with the operational guidance and rules of engagement of both sides, determine the possible operational threats in various directions and fields of battle operations. For the tactical level, based on the quality and performance of weapons and equipment of both sides, we study the typical battlefield confrontation environment and winning strategies, and identify the engagement threats faced in specific spatial and temporal engagement scenarios.

Categorize the mission. Based on the national security and development strategy, national defense strategy, military strategy, etc., combined with global political, economic, military, and diplomatic situation changes, study and analyze the overall war concept, typical operational scenarios and specific combat scenarios, and confirm the mission tasks of military subjects at different levels. For the strategic level, we will comprehensively consider the strategic threats and challenges faced, combine the current trends in the development of military power and strategic deployments of both sides, and generally position the future long-term core strategic missions and the overall concept of future warfare. For the battle level, comprehensive consideration of the possible operational threats in each direction and field, combined with the layout of the military force battlefield, decomposition of the future medium and long-term operational concept and the main operational tasks. For the tactical level, according to the battlefield threats faced by a specific time and space engagement, combined with the use of combat forces and weapons and equipment development, refining the engagement scenarios and specific combat operations in different time and space.

The problem of military problems. According to the degree of impact of threats and challenges, the main contradictions and key needs of military problems are grasped, and a list of military problems that need to be solved by military subjects at different levels is formed. For the strategic level, focusing on the strategic game, to affect the overall situation of the strategy, determine the success or failure of the war, involving long-term development as the point, around the seizure of the initiative in war, sorting out the list of issues to carry out strategic missions. For the battle operation, to affect the overall situation of the battle, determine the battlefield victory or defeat, involving the system development as the standard, around the seizure of the battlefield control, sorting out the formation of combat mission of the list of issues. Focus on tactical confrontation, to affect the battlefield engagement, determine the battlefield victory or defeat, involving the development of elements as the standard, around the seizure of combat control, sorting out the formation of a list of problems in attempting combat activities.

Adhere to the effect-oriented, select the optimal use of the program

Methodology and strategy is the core fulcrum of operational concept development. The core work of methodological strategy design is to adhere to the effect of optimal energy efficiency orientation, analysis and assessment of capability gaps, custom design strategies and methods of warfare, the best selection of the best options, and the utility of the operational concept development off.

Find the breaking point. For strategic, campaign, tactical different subjects to carry out the task facing the threat conditions, corresponding analysis of the war system, combat system, operational design on the two sides of the current analysis of military power, in the dynamic integrated game to seek symmetric, asymmetric or both breakthrough point of victory. For strategic adversaries, macro analysis of the "total potential difference" in the comprehensive strength of the war system between the two sides, based on the overall victory, from the perspective of the overall planning and use of war resources to seek a breakthrough point to win the strategic initiative. For the combat system, analysis and measurement of the two sides of the combat system "capacity difference", based on the system to win, from the perspective of the planning and use of combat forces to seek to win the initiative of the entry point. For the warring opponents weapons and equipment entities, directional analysis of the two sides against the nodes of the "technical difference", based on means of winning, from the perspective of weapons and technology integrated planning and use to seek to win the tactical initiative of the starting point.

Set strategy. According to the current capability gaps and future technological development possibilities, deep study of the inner laws of winning future intelligent warfare, based on the use of existing forces and new weapons development, proactive design, customizing different levels of the solution strategy set. For the strategic level, focus on the laws of war, focus on "what war to fight", build an agile anti-destructive network as the core, research and propose strategic guidance and overall tactics set under different war concepts, forming a top-level concept. For the campaign level, focus on the study of combat winning mechanism, around the "how to fight the war", to penetrate the land, sea, air and sky network kill chain as the core, research and propose different operational concepts under the set of operational methods of warfare, the formation of the concept of action. For the tactical level, focus on the study of the rules of engagement, targeting "how to protect", to support the kill chain "detection - control - resistance - fight - evaluation "nodes as the core, research and propose different engagement scenarios under the set of operational processes, the formation of support concepts.

Solution Selection. Based on various dimensions, the strategy ensemble is evaluated and analyzed to select the best solution that is consistent with the national military strategic objectives and has good support and realization possibilities under the constraints of technology, economy, and time. For the top-level concept, the best solution is selected according to the matching degree of military strategy, the support degree of core mission tasks, and the traction degree of military force development; for the operation concept, the best

solution is selected according to the support degree of major combat tasks and the enhancement degree of system capability; for the support concept, the best solution is selected according to the matching degree of equipment technology development, the support degree of major engagement activities, and the efficiency-cost ratio. For the support concept, the focus is on the degree of matching with the development of equipment technology, the degree of support for the main engagement activities, and the best program according to the efficiency ratio.

Adhere to the goal-oriented, set the actual construction needs

The concept of combat is based on the support of the development of the concept. The core work of the demonstration of capability support is to adhere to the goal-oriented grasp of building for warfare, stratified by the construction needs of the domain research, unified and refined capability indicators, design development roadmap and construction drawings, to keep the support of the development of the concept of warfare.

Research and propose requirements. Refine different levels of capability requirements for different levels of operational concepts and targeting. For the top-level concept, focus on warfare capability requirements, decompose and form the overall development requirements of the country and the military in different areas at different times. For the operational concept, focus on the operational capability requirements, around the force, equipment, training and other aspects, research and propose the system construction requirements in various fields and directions. For the support concept, focus on the requirements of the engagement function, refine the equipment and technology construction requirements to directly support the engagement.

Consolidation of indicators. According to the capability requirements, combine the existing system construction and medium and long-term construction and development planning, refine the capability indicators of corresponding levels, and unify to form the echelon construction indicators. In response to the top-level concept capability needs, in accordance with the overall development requirements, the top-level national and military indicators on the quantitative scale of war resources and strategic capability level construction are formed. For the operational concept capability needs, in accordance with the system construction requirements, the quantitative indicators of the scale structure and operational capability level of the combat system in each field and direction are formed. For the support concept capability needs, according to the equipment and technology construction needs, combined to form the quantitative indicators of the scale structure of the means of combat and core engagement functions.

Planning Development. Design development plans for combat forces, weapons and equipment, and key technologies in a progressive manner, taking as basic constraints the degree of combat

effectiveness satisfaction, development time frame adequacy, resource security, and technology maturity. For the top-level concept capability indicators, around the establishment of national and military development strategies, consider strategic relevance, development sustainability, etc., and design the overall construction blueprint to effectively support mission tasks. For the operational concept capability indicators, around the development of medium- and long-term construction planning, taking into account operational urgency, capability continuity, etc., design the roadmap for the construction of capabilities in various fields and directions to effectively support operational tasks. For the support concept capability indicators, around the development of a certain stage of the construction plan, taking into account the intensity of the engagement, capability support, etc., to design the construction of equipment technology development to effectively support the engagement activities.

(Author's unit: Rocket Research Institute)

“研究军事、研究战争、研究打仗”专论

作战概念究竟如何开发

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引言

作战概念是前瞻设计未来战争、引领军队建设备战的重要理论抓手，它源于战争准备、并受战争实践检验。从本质上看，作战概念开发是基于战争形态演变、前沿技术发展、作战对手和对抗环境的变化，针对战略、战役、战术不同层次的作战问题，对方法策略和能力支撑方案的设计优化，进而“作战问题+方法策略+能力支撑”的基本范式。

坚持效果导向，选准最优运用方案

坚持目标导向，定准实际建设需求

群策集

战略规划实施须重视

■刘玲成